A Macroeconomic Model of the Cross-section of Currencies

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motivation

What drives the cross-sectional variation of bilateral exchange rates?

International finance approach: Verdelhan (2018), Lustig and Richmond (2020), ...

- two factors account for most of variation, countries heterogeneously exposed to factors

International macro advances: Gabaix and Maggiori (2015), Itskhoki and Mukhin (2021), ...

- models connect exchange rate variation to macro, focus on single exchange rate (dollar)

This paper:

- macro model that explains factor structure of exchange rates & their drivers

what we do

Model of the global economy with a cross-section of currencies:

- heterogeneous countries: dollar assets, exposure to commodity cycle
- segmented asset markets & global financial intermediaries
- shocks: finanicial vs real, local vs global
- exchange rates have factor structure: "dollar" and "commodity" factors
- dollar factor = US output + risk shocks, commodity factor = global output + risk shocks

Empirical and quantitative analysis:

- the two factors explain around 50% of variation in exchange rates in data
- factor loadings in data explained by dollar assets and commodity intensity

Drivers of exchange rates: <u>no single answer</u>, depend on commodity exposure and dollar assets

related literature

- **factor structure in exchange rates and currency risk**:

  Lustig Verdelhan (07), Lustig Roussanov Verdelhan (11, 14), Menkhoff Sarno Schmeling Schrimpf (12), Hassan (13), Maggiori (17), Ready Roussanov Ward (17), Verdelhan (18), Colacito Croce Gavazzoni Ward (18), Kalemli-Özcan (19, 21), Richmond (19), Lustig Richmond (20), Koijen Yogo (20), Hassan Zhang (21), ...

- **macro models of exchange rates**:

  Stockman Tesar (95), Engel West (05), Colacito Croce (11), Schmitt-Grohe Uribe (17), Itskhoki Mukhin (21, 25), Chahrour Cormun De Leo Gerron Quintana Valchev (24), Engel and Wu (24,25), Jiang Krishnamurthy Lustig (24), Kekre Lenel (24,25), Bodenstein et al (24,25), ...

- **exchange rates and asset market imperfections**:

  Backus Smith (93), Lettau et al (14), Gabaix Maggiori (15), Lustig Verdelhan (19), Maggiori (22), Jiang Krishnamurthy Lustig (23), Chernov Haddad Istkhoki (24), Dao Gourinchas Itskhoki (25), ...

Model

households' problem

Continuum of small open economies indexed by $i \in [0,1]$, US is a mass point

Each economy: household + central bank

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u\left(C_{it}\right)$$

$$P_{it}^N C_{it}^N + C_{it}^T = P_{it}^N Y_{it}^N + Y_{it}^T + R_{it} Q_{it} B_{it} - Q_{it} B_{i,t+1} + T_{it} + \Pi_{it}$$

- save in local currency bond $B_{i,t+1}$ at R_{it}, get fiscal rebate T_{it}, profits of intermediaries Π_{it}
- P_{it}^N price of non-traded good
- $Q_{it} = (P_{it}^N)^\alpha$ price of consumption basket $C_{it} \propto (C_{it}^N)^\alpha (C_{it}^T)^{1-\alpha}$
- $S_{ijt} = \dfrac{Q_{jt}}{Q_{it}} = \left(\dfrac{P_{jt}}{P_{it}}\right)^\alpha$ real exchange rate

households endowments & optimality conditions

Endowments:

- non-tradable endowment: $Y_{it}^{N} = N(1 + x_{it})$, idiosyncratic shocks x_{it}
- tradable endowment: $Y_{it}^{T} = 1 + e_i z_t$: country-specific exposure e_i to global shock z_t

Exposure to global shock $e_i \sim$ commodity incidence. Microfoundation:

- tradable goods produced with commodities and labor
- heterogeneous endowments of commodities across countries
- commodity-biased productivity shocks map to z_t, commodity endowments to e_i

central banks

Buy foreign reserves $M_{i,t+1}$ (dollar bonds), transfers proceeds to households:

$$T_{it} = R_{ut} Q_{ut} M_{it} - Q_{ut} M_{i,t+1}$$

Reserve policy:

$$M_{i,t+1} = M_i \cdot \left(1 + \tau \left(\frac{Q_t}{Q_{ut}} - 1\right)\right)$$

- exogenous heterogeneity in reserve level M_i
- captures domestic policy uncertainty (eg, Bianchi Hatchondo Martinez (18), Drenik Kirpalani Perez (22))
- price of US consumption basket Q_{ut}, US interest rate R_{ut}, average price $Q_t \equiv \int_0^1 Q_{it} \mathrm{d}i$
- sell reserves when dollar strong $(0 < \tau)$

US problem: same as country i, no reserves (wlog)

country equations

Optimality conditions:

$$
\begin{aligned}
\textbf{intra-temporal condition:} \quad && P^N_{it} =& \frac{\alpha}{1-\alpha}\frac{C^T_{it}}{C^N_{it}} \\
\textbf{euler equation:} \quad && 1 =& \beta R_{i,t+1}\mathbb{E}_t\left[\frac{u'(C_{i,t+1})}{u'(C_{it})}\right]
\end{aligned}
$$

Country resource constraint:

$$
C^T_{it} = Y^T_{it} + Q_{it}(R_{it}B_{it} - B_{i,t+1}) + Q_{ut}(R_{ut}M_{it} - M_{i,t+1}) + \Pi_{it}
$$

global financial intermediaries

Representative intermediary invests in currency i bonds $A_{i,t+1}$ for all i, including US:

- zero-capital investment: $\int Q_{it} A_{i,t+1} \mathrm{d}i + Q_{ut} A_{u,t+1} = 0$

Problem:

$$\max_{\{A_{i,t+1}\}_i} \mathbb{E}_t \left[\Lambda_{t+1} \int Q_{it} A_{i,t+1} \Big(\underbrace{R_{i,t+1} \frac{Q_{i,t+1}}{Q_{it}} - R_{u,t+1} \frac{Q_{u,t+1}}{Q_{ut}}}_{\text{excess return vs USD}(\equiv X_{i,t+1})} \Big) \mathrm{d}i \right] - \underbrace{\frac{1}{2\chi} \int \left(Q_{it} A_{i,t+1} - L_{t+1} \right)^2 di}_{\text{portfolio cost}}$$

- benchmark level $L_{t+1} = Q_{ut} \int M_{i,t+1} \mathrm{d}i$ set so intermediates absorb dollar reserves demand
- optimal portfolio:

$$Q_{it} A_{i,t+1} = L_{t+1} + \chi \left(\mathbb{E}_t[\Lambda_{t+1}] \mathbb{E}_t[X_{i,t+1}] + \mathbb{C}_t \left[X_{i,t+1}, \Lambda_{t+1} \right] \right)$$

intermediaries' sdf

Intermediaries' SDF is a mean-preserving spread of households' SDF:

$$\Lambda_{t+1} = \Gamma_t \cdot \underbrace{\left[\int \Lambda_{i,t+1} \mathrm{d}i + \Lambda_{u,t+1}\right]}_{\text{average SDF}} + (1 - \Gamma_t) \cdot \underbrace{\mathbb{E}_t \left[\int \Lambda_{i,t+1} \mathrm{d}i + \Lambda_{u,t+1}\right]}_{\text{expected average SDF}}$$

Households' SDF given by marginal utility of tradables:

$$\Lambda^T_{i,t+1} \equiv \beta \frac{\partial u(C_{i,t+1})}{\partial C^T_{i,t+1}} \left(\frac{\partial u(C_{i,t})}{\partial C^T_{i,t}} \right)^{-1}$$

- $\Gamma_t = \Gamma(1 + \gamma_t)$ aggregate shock to intermediaries' risk aversion

equilibrium

Equilibrium is a set of processes $\{P_{it}^N, Q_{it}, R_{it}\}$ and $\{C_{it}^N, C_{it}^T, B_{i,t+1}, A_{i,t+1}, M_{i,t+1}\}$ such that the quantities solve the optimization problems and the following markets clear:

$$
\begin{aligned}
&\text{non-tradable goods } i: && C_{it}^N = Y_{it}^N \\
&\text{tradable goods:} && \int C_{it}^T \mathrm{d}i + C_{ut}^T = \int Y_{it}^T \mathrm{d}i + Y_{ut}^T \\
&\text{currency } i \text{ bonds:} && A_{i,t+1} + B_{i,t+1} = 0 \\
&\text{dollar bonds:} && \int \underbrace{M_{i,t+1} di}_{\text{reserves}} \underbrace{+ A_{u,t+1}}_{\text{intermediary}} \underbrace{+ B_{u,t+1}}_{\text{US HH}} = 0
\end{aligned}
$$

Shocks: $(z_t, x_{ut}, \gamma_t, \{x_{it}\})$ all AR-1

Solving the model

solution

Log-linear approximation around steady state, limit of small shocks & large risk aversion

- simultaneously $(\sigma_z, \sigma_u, \sigma_\gamma) \longrightarrow 0$ and $(\Gamma\sigma_z^2, \Gamma\sigma_u^2, \Gamma\sigma_\gamma^2) \longrightarrow (\Gamma_z, \Gamma_u, \Gamma_\gamma)$
- covariance between SDF and exchange rates survives in the double limit

Intermediaries' SDF:

$$\lambda_{t+1}^{\text{ave}} = -\lambda_z \Delta z_{t+1} - (\rho - 1)\lambda_u \Delta x_{u,t+1}$$

- $\lambda_z > 0$: global marginal utility of tradables $\uparrow$ when global supply $\downarrow$
- $\lambda_u > 0$: US marginal utility of tradables $\uparrow$ when non-traded consumption $\downarrow$ <u>(if IES<1)</u>

exchange rates and capital flows

Exchange rate determined by endowments, inflows $\Delta l_{i,t+1}$, and reserve accumulation $\Delta m_{i,t+1}$:

$$q_{it} = \alpha(\overbrace{e_i z_t + \Delta l_{i,t+1} - M_i \Delta m_{i,t+1}}^{\text{tradable consumption}}) - \alpha x_{it}$$

Capital inflows:

Euler equations | reserve accumulation

$$l_{i,t+1} = \underbrace{\chi \mathbb{E}_t[\Delta q_{i,t+1} - \Delta q_{u,t+1} + r_{i,t+1} - r_{u,t+1}]}_{\text{expected excess returns}} + \underbrace{\gamma_t \chi \mathcal{C}_i}_{\text{rp shock}} + \underbrace{L \Delta m_{t+1}}_{\text{expansion}}$$

Limiting risk premium:

$$\mathcal{C}_i = \lim_{(\sigma_z, \sigma_u, \sigma_\gamma) \longrightarrow 0,\ \Gamma \longrightarrow \infty} \mathbb{C}_t[\lambda_{t+1}^{\text{ave}}, \Delta q_{i,t+1} - \Delta q_{u,t+1}]$$

solving for exchange rates

Tractable equilibrium where $\rho_x = \rho_z = 1$, $\rho_\gamma = 0$, and $\chi \longrightarrow \infty$ with $\chi^{-\frac{1}{2}} M_i \longrightarrow m_i$

Changes in bilateral exchange rate vis-a-vis US: $\Delta s_{i,t+1} \equiv \Delta q_{u,t+1} - \Delta q_{i,t+1}$

$$\Delta s_{i,t+1} = (e_u - e_i) \cdot \alpha \Delta z_{t+1} + (\tau\phi(m_i - m) - 1) \cdot \alpha \Delta x_{u,t+1} + \kappa_i \cdot \Delta\gamma_{t+1} + \alpha \Delta x_{i,t+1}$$

- exposure to z_t depends on e_i vs e_u
- exposure to x_{ut} reflects two effects: direct from goods market + indirect from assets
- -1 reflects non-tradable expansion in US
- $\phi\tau(m_i - m)$ reflects reserve accumulation net of recycled capital inflows in country i
- exposure to γ_t is given by

$$\kappa_i = (e_i - e_u) \cdot (e + e_u)\alpha^2 \Gamma_z + (1 - \tau\phi(m_i - m)) \cdot \alpha\phi^2 \Gamma_u (\rho - 1)$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

$$\Delta s_{i,t+1} = \alpha \Delta x_{i,t+1} + \alpha(e_u - e_i) \cdot \underbrace{[\Delta z_{t+1} - (e + e_u)\alpha\Gamma_z^2 \Delta\gamma_{t+1}]}_{\text{commodity + commodity risk}}$$

$$+ \alpha(\phi\tau(m_i - m) - 1) \cdot \underbrace{[\Delta x_{u,t+1} - (\rho - 1)\phi^2\Gamma_u^2 \Delta\gamma_{t+1}]}_{\text{US output + US output risk}}$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

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Dollar factor isolates the US-related part:

$$\text{doll}_{t+1} \equiv \int \Delta s_{i,t+1} \mathrm{d}i$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

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<u>Dollar factor</u> isolates the US-related part:

$$\text{doll}_{t+1} = \underbrace{\int \alpha\Delta x_{i,t+1}\mathrm{d}i}_{=0} + [\Delta z_{t+1} - (e + e_u)\alpha\Gamma_z^2\Delta\gamma_{t+1}] \cdot \underbrace{\int \alpha(e_u - e_i)\mathrm{d}i}_{=0 \text{ if } e_u = e}$$

$$+ [\Delta x_{u,t+1} - (\rho - 1)\phi^2\Gamma_u^2\Delta\gamma_{t+1}] \cdot \underbrace{\int \alpha(\phi\tau(m_i - m) - 1)\mathrm{d}i}_{=-\alpha}$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

$$\Delta s_{i,t+1} = \alpha \Delta x_{i,t+1} + \alpha(e_u - e_i) \cdot \underbrace{[\Delta z_{t+1} - (e + e_u)\alpha \Gamma_z^2 \Delta \gamma_{t+1}]}_{\text{commodity + commodity risk}}$$

$$+ \alpha(\phi\tau(m_i - m) - 1) \cdot \underbrace{[\Delta x_{u,t+1} - (\rho - 1)\phi^2 \Gamma_u^2 \Delta \gamma_{t+1}]}_{\text{US output + US output risk}}$$

Dollar factor isolates the US-related part:

$$\text{doll}_{t+1} = -\alpha \cdot [\Delta x_{u,t+1} - (\rho - 1)\phi^2 \Gamma_u^2 \Delta \gamma_{t+1}]$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

$$\Delta s_{i,t+1} = \alpha \Delta x_{i,t+1} + \alpha(e_u - e_i) \cdot \underbrace{[\Delta z_{t+1} - (e + e_u)\alpha \Gamma_z^2 \Delta \gamma_{t+1}]}_{\text{commodity + commodity risk}}$$

$$+ \alpha(\phi\tau(m_i - m) - 1) \cdot \underbrace{[\Delta x_{u,t+1} - (\rho - 1)\phi^2 \Gamma_u^2 \Delta \gamma_{t+1}]}_{\text{US output + US output risk}}$$

Commodity factor isolates the commodity part:

$$\text{comm}_{t+1} \equiv \int_{i:e_i > e} 2\Delta s_{i,t+1} \mathrm{d}i - \int_{i:e_i < e} 2\Delta s_{i,t+1} \mathrm{d}i$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

$$\Delta s_{i,t+1} = \alpha\Delta x_{i,t+1} + \alpha(e_u - e_i) \cdot \underbrace{[\Delta z_{t+1} - (e + e_u)\alpha\Gamma_z^2 \Delta\gamma_{t+1}]}_{\text{commodity + commodity risk}}$$

$$+ \alpha(\phi\tau(m_i - m) - 1) \cdot \underbrace{[\Delta x_{u,t+1} - (\rho - 1)\phi^2\Gamma_u^2 \Delta\gamma_{t+1}]}_{\text{US output + US output risk}}$$

<u>Commodity factor</u> isolates the commodity part (assuming $e_i \perp M_i$):

$$\text{comm}_{t+1} = \alpha(e_L - e_H) \cdot [\Delta z_{t+1} - (e + e_u)\alpha\Gamma_z^2 \Delta\gamma_{t+1}]$$

factor structure of exchange rates

Group terms into commodity-related and US-related:

$$\Delta s_{i,t+1} = \alpha \Delta x_{i,t+1} + \alpha(e_u - e_i) \cdot \underbrace{[\Delta z_{t+1} - (e + e_u)\alpha\Gamma_z^2 \Delta\gamma_{t+1}]}_{\text{commodity + commodity risk}}$$

$$+ \alpha(\phi\tau(m_i - m) - 1) \cdot \underbrace{[\Delta x_{u,t+1} - (\rho - 1)\phi^2\Gamma_u^2 \Delta\gamma_{t+1}]}_{\text{US output + US output risk}}$$

Plug factors:

$$\boxed{\Delta s_{i,t+1} = \frac{e_i - e}{e_H - e_L} \cdot \text{comm}_{t+1} + (1 - \phi\tau(m_i - m)) \cdot \text{doll}_{t+1} + \epsilon_{i,t+1}}$$

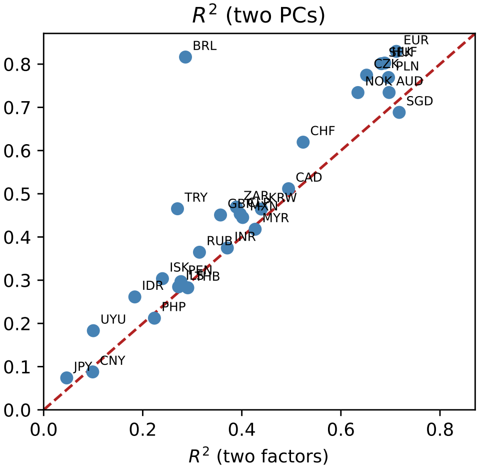
Empirical analysis

factor regression

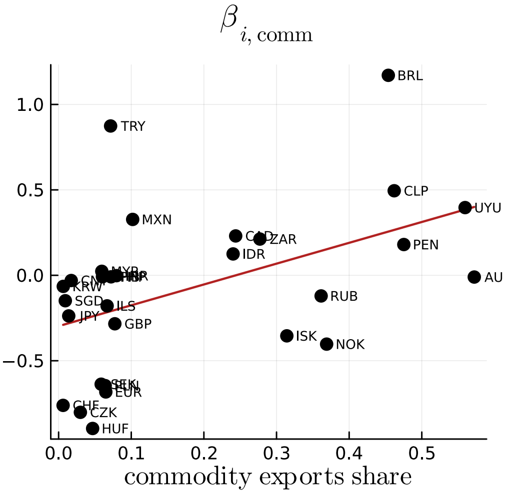
Run the same regression:

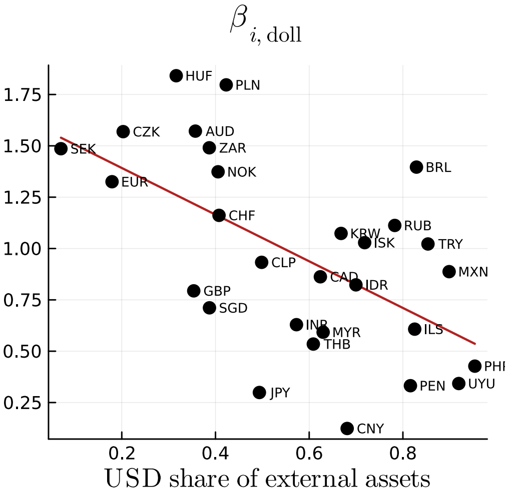
$$\Delta s_{i,t+1} = \beta_{i,\text{comm}} \cdot \text{comm}_{t+1} + \beta_{i,\text{doll}} \cdot \text{doll}_{t+1} + \varepsilon_{i,t+1}$$

commodity loadings | dollar loadings | explanatory power



determinants of loadings





more evidence on the mechanisms

Factors explained by empirical measures of shocks

- doll_t $\searrow$ in nontradable employment & $\nearrow$ in risk measures (VIX, EBP, intermediate leverage)
- comm_t $\searrow$ in commodity prices & $\nearrow$ in risk measures
- comm_t $\sim$ similar variation as trade centrality & carry factors Verdelhan (18), Richmond (19) more

Heterogeneous exposure to the dollar factor due to realized capital gains

- relevance of asset accumulation: net acquisition of USD assets vs consumption magnitudes

additional empirical results

Factor-structure of exchange rates in another base j

- through the lens of the model: 3-factor structure $(\text{doll}_t, \text{comm}_t, \text{base}(j)_t)$ base j factor
- data: doll_t remains significant in $> \frac{3}{4}$ of the regressions for all possible bases EUR and GBP bases

Role of comm_t & including other cross-country controls

- same results if replace commodity incidence with tradable output sensitivity loadings R^2
- controls: size, trade centrality, openness, gravity vars Hassan (13), Lustig Richmond (20) loadings

Quantitative analysis

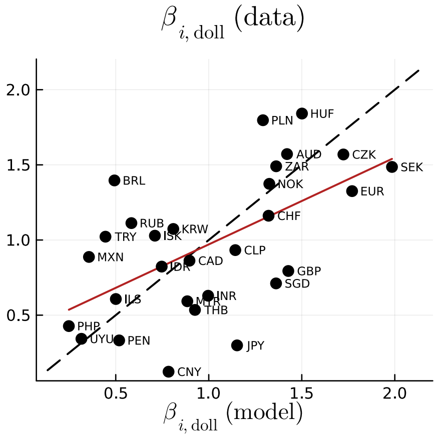
calibration

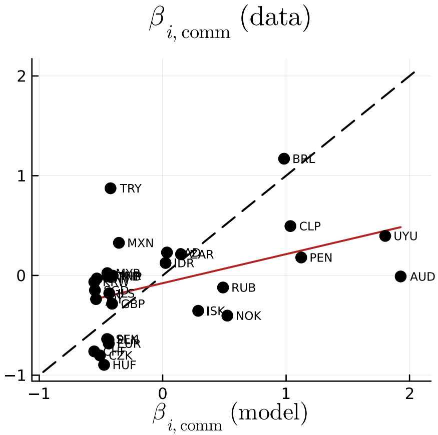
Target the three moments of factors + avg R^2 + xr volatility + xs dispersion of doll loadings

Moment	Data	Model
std(doll_t)	1.94pp	1.94pp
std(comm_t)	1.32pp	1.32pp
corr(doll_t,comm_t)	0.31	0.31
std(ε_{it})	2.48pp	2.48pp
std(Δs_{it})	3.24pp	3.34pp
avg R^2 of factors	41%	40%
std$(\beta_{i,\text{doll}})$	0.47	0.47
std$(\beta_{i,\text{comm}})$	0.48	0.73

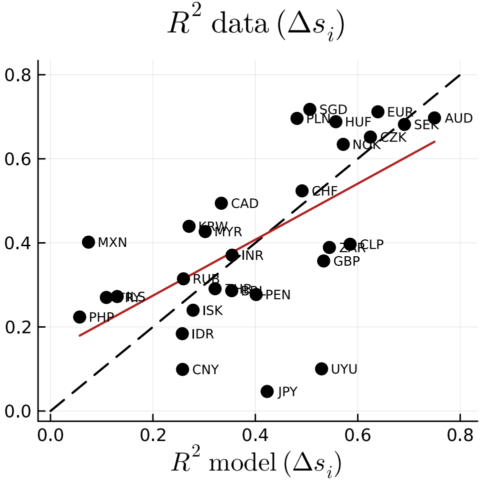
identification from factor moments

untargeted moments: country loadings on factors





heterogeneity in global vs idiosyncratic shocks



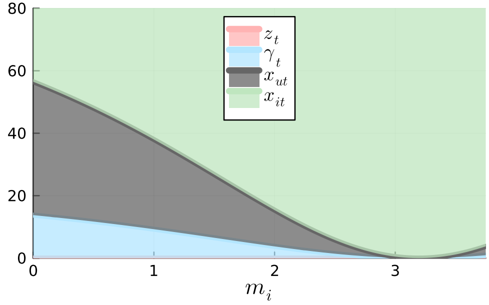
variance decomposition of US price level and factors

	Δq_{ut}	doll_t	comm_t
$\Delta \gamma_t$	7%	24%	40%
Δz_t	10%		60%
Δx_{ut}	84%	76%	

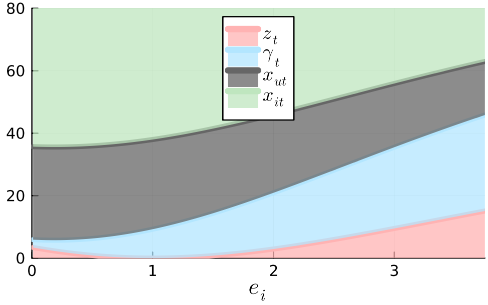
Financial shocks more important for dollar factor than for Δq_{ut}

- financial shocks are redistributive: move capital flows frow RoW to US

drivers of exchange rates (vis-a-vis dollar) in cross-section



(a) shares of exchange rate variance, $e_i = e$



(b) shares of exchange rate variance, $m_i = m$

- US non-tradable & financial shocks matter more for countries with low dollar assets
- financial shocks matter more for commodity producers
- global shocks as important as idiosyncratic shocks on average

drivers of bilateral exchange rates vis-a-vis dollar

		Thailand	Hungary	Australia
(a) Data				
std(Δs_{it})		2.0%	3.9%	3.6%
factor R^2		29%	69%	70%
e_i		low	low	high
m_i		high	low	low
(b) Model				
std(Δs_{it})		3.0%	3.7%	5.0%
factor R^2		32%	56%	75%
Δs_{it} variance shares	$\Delta \gamma_t$	9%	14%	48%
	Δz_t	7%	3%	21%
	Δx_{ut}	84%	83%	31%

conclusion

Macroeconomic model:

- output + financial shocks $\mapsto$ dollar and commodity factors
- heterogeneity in exposure to commodities and dollar assets $\mapsto$ loadings on factors

Replicate factors and loadings in the model

Decompose variation in exchange rates into fundamental sources

- financial shocks contribute <u>more</u> to dollar factor than to US prices
- financial shocks and US shocks consistently important
- high dollar assets insulate currencies from global shocks

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Euler equations and Backus-Smith

Consumption and exchange rate are related:

$$c_{it} = \frac{1-\alpha}{\alpha} q_{it} + x_{it}$$

Backus-Smith covariance:

$$\mathbb{C}[c_{it} - c_{ut}, q_{it} - q_{ut}] = \frac{1-\alpha}{\alpha} \mathbb{V}[q_{it} - q_{ut}] + \mathbb{C}[q_{it} - q_{ut}, x_{it} - x_{ut}]$$

Euler equation leads to

$$r_{it} = \frac{1-\alpha}{\alpha} \mathbb{E}_t[\Delta q_{i,t+1}] + \mathbb{E}_t[\Delta x_{i,t+1}]$$

back

reserve accumulation

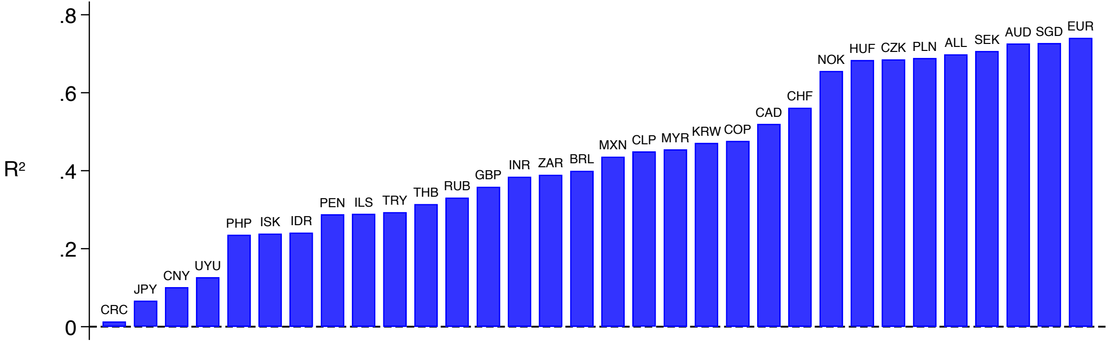
Reserve accumulation symmetric in logs, positive when dollar is cheap:

$$m_{i,t+1} \equiv m_{t+1} = \tau(q_t - q_{ut})$$

back

factor regressions: explanatory power

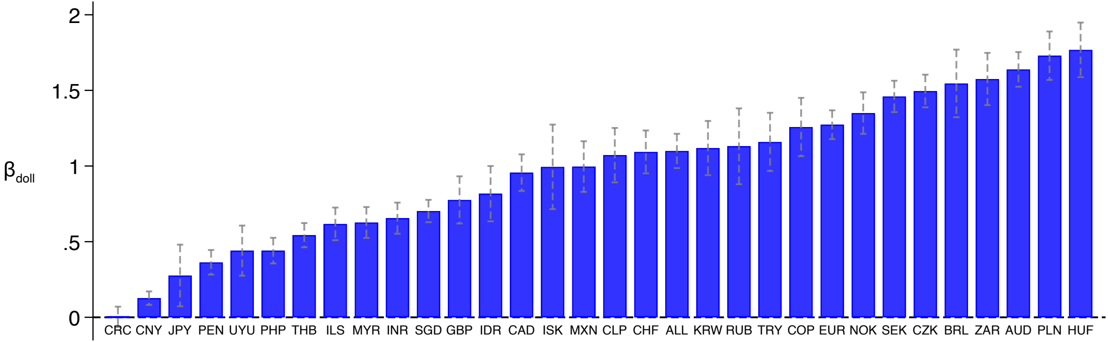
$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$



back

factor regressions: dollar loadings

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$



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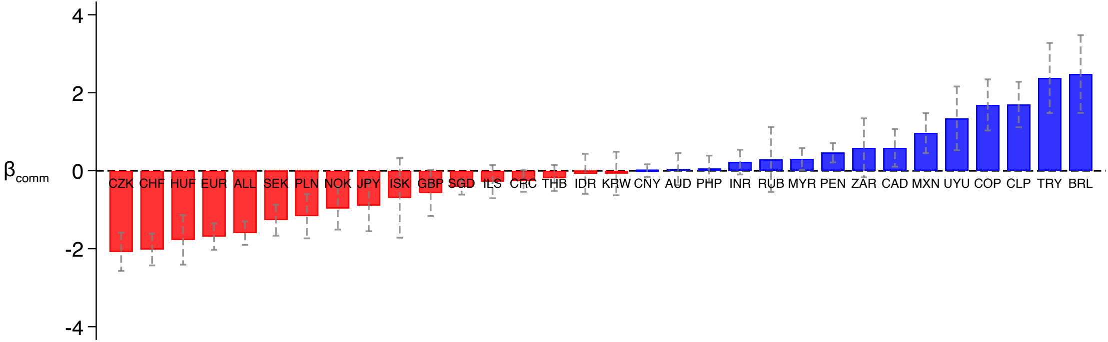
factor regressions: commodity loadings

$$\Delta s_{i,t+1} = \beta_{i,\text{comm}}\text{comm}_{t+1} + \beta_{i,\text{doll}}\text{doll}_{t+1} + \varepsilon_{i,t+1}$$



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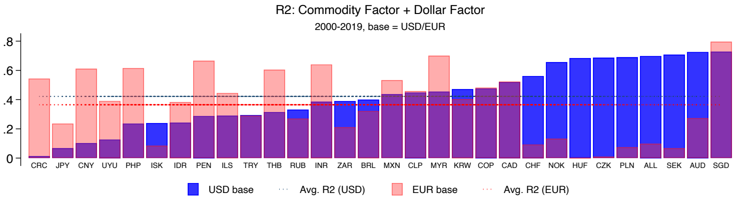
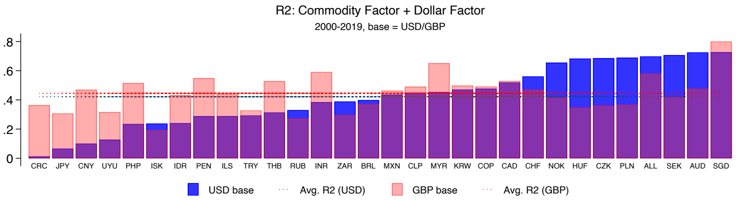
explanatory power for EUR/GBP bases





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carry explanatory power

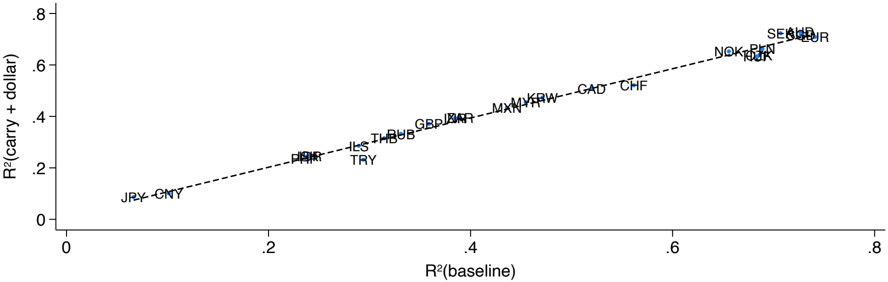

Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the carry factor

back

carry loadings

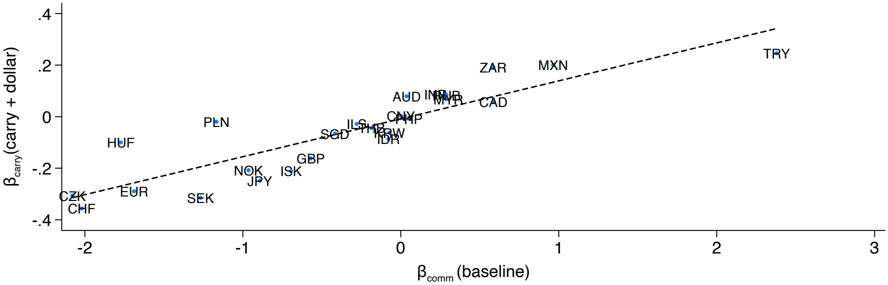

Figure: loadings on the carry factor vs commodity factor, bivariate regressions that include the dollar factor

back

net acquisition of USD assets

Ratio of std of net USD asset acquisition and consumption

- median: 0.24
- average: 0.34

Time-series correlation of net USD asset acquisition and consumption

- median: -0.28
- average: -0.27

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trade centrality explanatory power

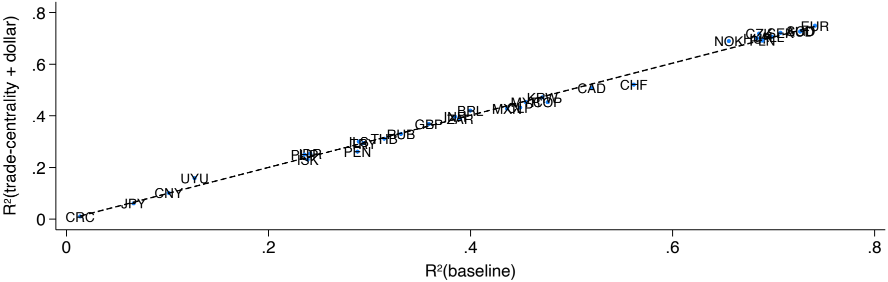

Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the trade centrality factor

back

trade centrality loadings

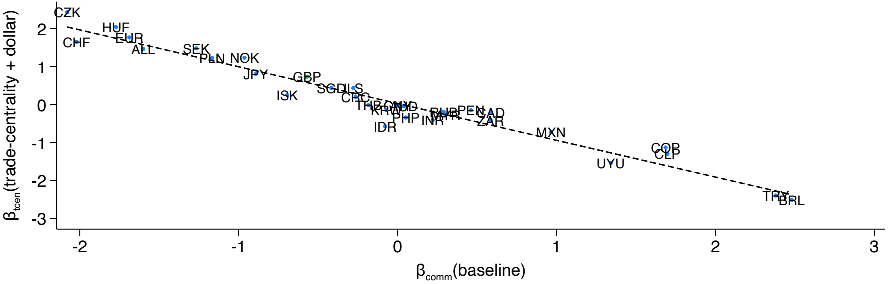

Figure: loadings on the trade centrality factor vs commodity factor, bivariate regressions that include the dollar factor

back

tradable output sensitivity explanatory power

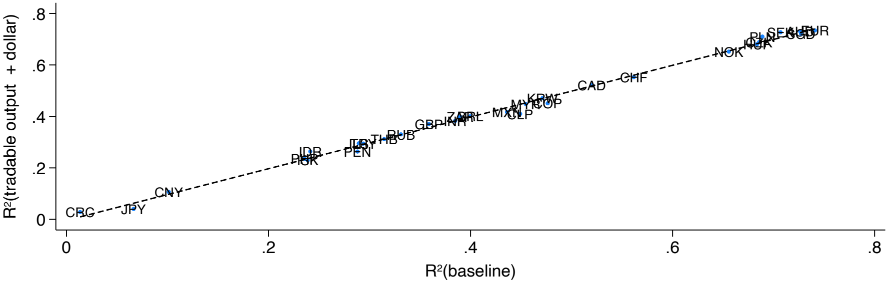

Figure: explanatory power of bivariate factor regressions with the dollar factor and either the commodity or the tradable output sensitivity factor

back

tradable output sensitivity loadings

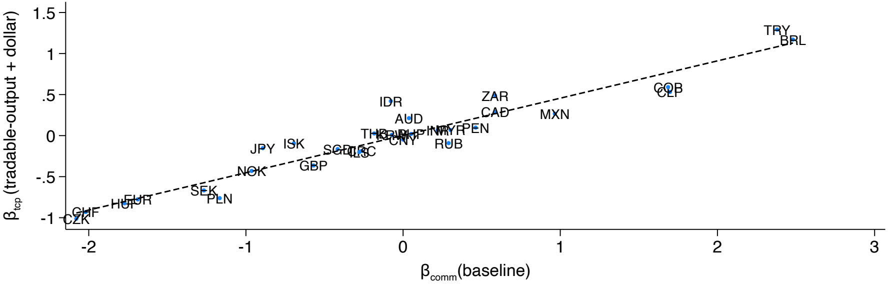

Figure: loadings on the tradable output sensitivity factor vs commodity factor, bivariate regressions that include the dollar factor

back

identification from factor moments

An invertible map $(\sigma_u^2, \sigma_z^2, (\Gamma\sigma_\gamma)^2) \mapsto (\mathbb{V}[\text{doll}_t], \mathbb{V}[\text{comm}_t], \mathbb{C}[\text{comm}_t, \text{doll}_t])$:

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$$\mathbb{V}[\text{doll}_t] = \sigma_u^2 + 2\lambda_u^2 \cdot \underbrace{\sigma_u^4(\Gamma\sigma_\gamma)^2}_{\text{pricing of US-specific risk}}$$

$$\mathbb{V}[\text{comm}_t] = \sigma_z^2 + 2\lambda_z^2 \cdot \underbrace{\sigma_z^4(\Gamma\sigma_\gamma)^2}_{\text{pricing of commodity risk}}$$

$$\mathbb{C}[\text{comm}_t, \text{doll}_t] = 4\lambda_z^2\lambda_u^2\sigma_z^2\sigma_u^2 \cdot (\Gamma\sigma_\gamma)^2$$

- both factors combine underlying shocks and shocks to their pricing
- factors are only correlated due to $\gamma_t \longrightarrow$ correlation informative on its strength
- variance informative on the strength of shocks to z_t and x_{ut}

risk-sharing moment

Risk sharing typically assessed by $\text{corr}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$ back

- equal to 1 with complete and integrated markets
- close to zero (negative) in the data

Proposition 5

The risk-sharing covariance $C_i = \mathbb{C}[\Delta c_{i,t+1} - \Delta c_{u,t+1}, \Delta s_{i,t+1}]$ is

$$C_i = F_x(m_i)\sigma_x^2 - F_z(e_i)\sigma_z^2 - F_\gamma(e_i, m_i)\sigma_\gamma^2$$

Here $F_z(\cdot) > 0$ and $F_\gamma(\cdot) > 0$.

- without shocks to x_{ut}, correlation always equal to -1

base-j factor structure

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$$\Delta s_{ij,t+1} = \frac{e_i - e}{e_H - e_L} \cdot \text{comm}_{t+1} - \tau\phi(m_i - m) \cdot \text{doll}_{t+1} + 1 \cdot \text{base}_{j,t+1} + \alpha \Delta x_{i,t+1}$$